

EXPERIMENT-01

1) Construct a

i) matrix of size m*n of your choice and

ii) vector of size m*n

Code:

>> A = [7 2 1; 4 8 2];

>> v = [5 9 3 7];

A

v

Output:

A =

7	2	1
4	8	2

>> v

v =

5	9	3	7
---	---	---	---

2) Construct a random matrix from

a) uniform distribution

b) normal distribution

c) integer from 1 to 100

d) real number from 0 to 1

Code:

>> m = input('Enter rows: ');

n = input('Enter columns: ');

U = rand(m,n); % Uniform random numbers (0 to 1)

N = randn(m,n); % Normal random numbers

I = randi([1 100],m,n); % Random integers from 1 to 100

R = rand(m,n); % Another random matrix

Output:

```
U
Enter rows: 3
Enter columns: 3|

U =

    0.8147    0.9134    0.2785
    0.9058    0.6324    0.5469
    0.1270    0.0975    0.9575

>> N

N =

    2.7694    0.7254   -0.2050
   -1.3499   -0.0631   -0.1241
    3.0349    0.7147    1.4897

>> I

I =

    80     4    68
    96    85    76
    66    94    75

>> R

R =

    0.3922    0.7060    0.0462
    0.6555    0.0318    0.0971
    0.1712    0.2769    0.8235
```

3) Find sum of

a) all rows

b) all columns

c) all entries of a matrix

Code:

>> A = input('Enter matrix: ');

sum_rows = sum(A,2);

sum_cols = sum(A,1);

sum_all = sum(A,'all');

A

sum_rows

sum_cols

sum_all

Output:

```
Enter matrix: [7 5 3; 2 8 4]
```

```
A =
```

```
    7    5    3
    2    8    4
```

```
|
```

```
sum_rows =
```

```
    15
```

```
    14
```

```
sum_cols =
```

```
    9    13    7
```

```
sum_all =
```

```
    29
```

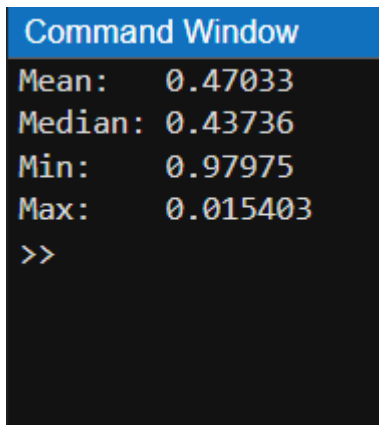
```
>>
```

- 4) Find
a) mean
b) median
c) min d) max

Code:

```
>> data = rand(1, 100);  
mean_val = mean(data);  
median_val = median(data);  
min_val = min(data);  
max_val = max(data);  
disp(['Mean: ' num2str(mean_val)])  
disp(['Median: ' num2str(median_val)])  
disp(['Min: ' num2str(max_val)])  
disp(['Max: ' num2str(min_val)])
```

Output:



Command Window

```
Mean:    0.47033  
Median:  0.43736  
Min:     0.97975  
Max:     0.015403  
>>
```

5) Find size of matrix and vector

Code:

>> clear

A = input('Enter matrix: ');

V = input('Enter vector: ');

size_A = size(A);

size_V = size(V);

size_A

size_V

Output:

```
Command Window
Enter matrix: [7 5; 9 3]
Enter vector: [10 55 46 37 14 79]|

size_A =

     2     2

size_V =

     1     6

>>
```

6) Find entries in a matrix whose entities are greater/smaller than particular number

Code:

>> A = input('Enter the matrix A: ');

n = input('Enter the number: ');

greaterEntries = A(A > n);

smallerEntries = A(A < n);

disp(greaterEntries)

disp(smallerEntries)

Output:

```
Command Window
Enter the matrix A: [43 99 21 34 91 79]
Enter the number: 50
    99    91    79

    43    21    34

>>
```

7) Find the inverse of a Matrix.

Code:

>> A = input('Enter the square matrix A: ');

if det(A) ~= 0

invA = inv(A);

disp(invA)

else

disp('Inverse does not exist')

end

Output:

Command Window

Enter the square matrix A: [91 97 34

21 37 55

75 68 46];

-0.0271 -0.0286 0.0542

0.0420 0.0218 -0.0571

-0.0179 0.0145 0.0177

>> |

8) Construct

a) diagonal matrix of size n

b) identity matrix of size n

c) zero matrix

Code:

>> d = input('Enter diagonal elements as a vector: ');

n = input('Enter size for identity/zero matrix: ');

D = diag(d);

I = eye(n);

Z = zeros(n);

disp('Diagonal Matrix:')

disp(D)

disp('Identity Matrix:')

disp(I)

disp('Zero Matrix:')

disp(Z)

Output:

Command Window

Enter diagonal elements as a vector: [7 3 1];

Enter size for identity/zero matrix: 3

Diagonal Matrix:

7	0	0
0	3	0
0	0	1

Identity Matrix:

1	0	0
0	1	0
0	0	1

Zero Matrix:

0	0	0
0	0	0
0	0	0

>>

9) Find product of two matrices

Code:

>> A = input('Enter first matrix: ');

B = input('Enter second matrix: ');

P = A * B;

P

Output:

```
Command Window
Enter first matrix: [7 6
1 3];|
Enter second matrix: [ 2 8
73 28];

P =

    452    224
    221     92

>>
```

10) Sort entries of matrices and vector in increasing and decreasing order.

Code:

>> V = input('Enter vector: ');

asc = sort(V);

desc = sort(V,'descend');

A = input('Enter matrix: ');

asc_M= sort(A);

desc_M= sort(A,'descend');

asc

desc

asc_M

desc_M

Output:

Command Window

Enter vector: [79 77 91 97 69 39 82 19 64 45];

Enter matrix: [32 78 16 51];

asc =

19	39	45	64	69	77	79	82	91	97
----	----	----	----	----	----	----	----	----	----

desc =

97	91	82	79	77	69	64	45	39	19
----	----	----	----	----	----	----	----	----	----

asc_M =

16	32	51	78
----	----	----	----

desc_M =

78	51	32	16
----	----	----	----

|

>>